YEAR 11 MATHS METHODS INVESTIGATIVE TASK 1 SOLUTIONS

Functions

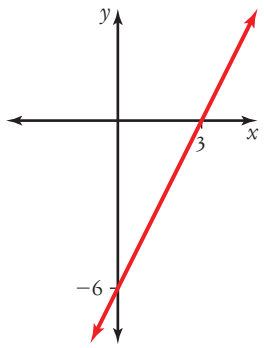
1 a i a straight line **[1]**

ii x- and y-intercepts, gradient **[2]**

iii x-intercept: $y = 0, 2x - 6 = 0 \Rightarrow x = 3$ **[1]**

y-intercept: $x = 0, y = 2 \times 0 - 6 = -6$ **[1]**

Using $y = mx + c$, gradient $= 2$ **[1]**



[2]

b i parabolic curve with a minimum value (concave upwards). **[2]**

ii x- and y-intercepts, turning point including if max or min

iii x-intercept: $y = 0, x^2 + 2x - 8 = 0 \Rightarrow (x - 2)(x + 4) = 0$

x-intercepts at $x = 2$ and $x = -4$ **[2]**

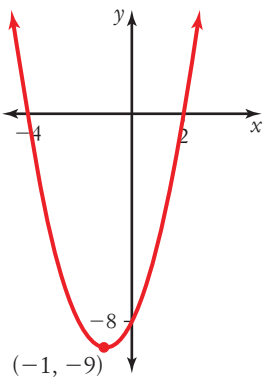
y-intercept: $x = 0, y = 0^2 + 2 \times 0 - 8 = -8$ **[1]**

Turning point: complete the square.

$y = x^2 + 2x + 1 - 8 - 1 = (x + 1)^2 - 9$ **[2]**

Turning point is $(-1, -9)$, **[1]**

It is a minimum as coefficient of x^2 is positive. **[1]**



[2]

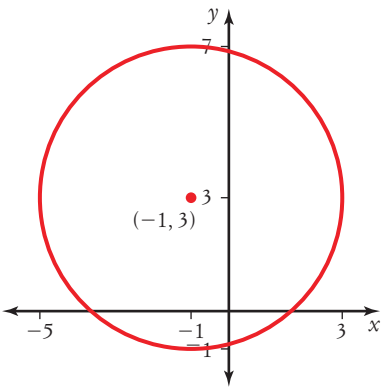
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c i a circle **[1]**

ii the centre and the radius **[1]**

iii centre $(-1, 3)$ and radius 4 **[2]**

It is helpful to locate the vertical and horizontal limits of the circle.



[2]

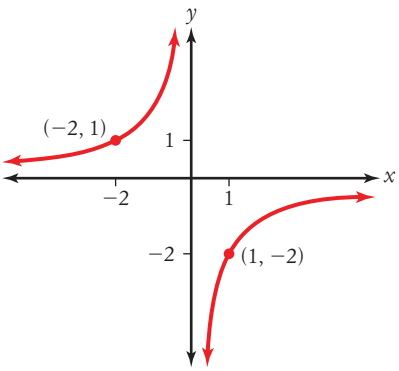
d i Change to $y = -\frac{2}{x}$. A rectangular hyperbola. **[2]**

ii x- and y-intercepts, asymptotes, quadrants **[2]**

iii The asymptotes will be the axes. **[1]**

No intercepts with axes, graph in 2nd & 4th quadrant. **[2]**

It is helpful to plot a point in each quadrant.



Other points may be plotted. **[2]**

e i A polynomial OR a quartic, a curve with both ends $\longrightarrow \infty$. **[1]**

ii x- and y-intercepts, behaviour as $x \longrightarrow \infty$ and $x \longrightarrow -\infty$ **[2]**

iii x-intercepts: $y = 0$, $(x - 2)^2(x + 1)(x + 3) = 0 \Rightarrow x = 2, -1, -3$ **[1]**

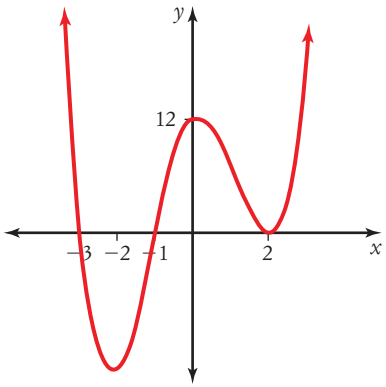
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Note: The first factor being a square may alert students to the graph touching the x-axis at this point.

y intercept: $x = 0$, $y = (-2)^2(1)(3) = 12$ **[1]**

Leading term is x^4, so as $x \longrightarrow \infty$, $y \longrightarrow \infty$ and as $x \longrightarrow -\infty$, $y \longrightarrow \infty$ **[1]**

Considering the sign of y for different values of x, the graph below is obtained. **[2]**



The Garden

2 The middle of the left seating is $\left(\frac{-2-9}{2}, \frac{1+9}{2}\right) = (-5.5, 5)$ **[1]**

The middle of the right seating is $\left(\frac{2+9}{2}, \frac{1+9}{2}\right) = (5.5, 5)$ **[1]**

The points are $(-9, 1)$, $(-5.5, 5)$, $(-2, 9)$, $(2, 9)$, $(5.5, 5)$, $(9, 1)$ **[1]**

3 Use the distance formula.

Length on left $= \sqrt{(-2-(-9))^2 + (9-1)^2} = \sqrt{7^2 + 8^2} = \sqrt{113} \approx 10.63$

The length is 10.63 m **[2]**

Length on right $= \sqrt{(2-9)^2 + (9-1)^2} = \sqrt{(-7)^2 + 8^2} = \sqrt{113} \approx 10.63$

All lengths must be at least 10.63 m **[1]**

Note: It is acceptable for students to state the second length is equal to the first from arguments using symmetry/reflection in y-axis.

4 a $A = \pi r^2 = \pi \times 2^2 \approx 12.6 \text{ m}^2$ so yes.

OR

$\pi r^2 \geq 10 \Rightarrow r^2 \geq \frac{10}{\pi} \Rightarrow r \geq 1.79$ so yes. **[1]**

b We can choose any centre for the circle that at least 2 m from any parts of the original design of the backyard, and more than 2 m from the y-axis.

Answers will vary but r^2 must be 4. Students should state the centre and radius of the chosen circle, and then write down its equation.

As an example, a student may choose a garden centre at $(8, -4)$ and radius 2, giving an equation of $(x-8)^2 + (y+4)^2 = 4$. **[3]**

5 a The seating, left side:

Gradient $= \frac{9-1}{-2-(-9)} = \frac{8}{7}$ **[1]**

EITHER Using $(-9, 1)$, $\frac{y-1}{x+9} = \frac{8}{7} \Rightarrow 7y - 7 = 8x + 72 \Rightarrow 8x - 7y + 79 = 0$

OR Using $(-2, 9)$, $\frac{y-9}{x+2} = \frac{8}{7} \Rightarrow 7y - 63 = 8x + 16 \Rightarrow 8x - 7y + 79 = 0$ **[2]**

Equation may be expressed as $y = \frac{8x}{7} + 11\frac{2}{7}$

Domain: $-9 \leq x \leq 2$ **[1]**

The seating, right side:

Gradient $= \frac{9-1}{2-9} = -\frac{8}{7}$ **[1]**

EITHER Using (2, 9), $\frac{y-9}{x-2} = -\frac{8}{7} \Rightarrow 7y - 63 = -8x + 16 \Rightarrow 8x + 7y - 79 = 0$

OR Using (9, 1), $\frac{y-1}{x-9} = -\frac{8}{7} \Rightarrow 7y - 7 = -8x + 72 \Rightarrow 8x + 7y - 79 = 0$ **[2]**

Equation may be expressed as $y = -\frac{8x}{7} + 11\frac{2}{7}$

Domain: $2 \leq x \leq 9$ **[1]**

b The fountain is a circle with centre the origin and radius 3 units. **[1]**

Equation: $x^2 + y^2 = 9$ **[2]**

c The curved sections of the northern garden beds:

Being a parabola, it will be of the form $y = ax^2 + bx + c$. **[1]**

x-intercepts of 10 and -10 show it must be $y = a(x - 10)(x + 10)$ **[1]**

A y-intercept of 10 means that when $x = 0$, $y = 10$ **[1]**

$10 = a \times -10 \times 10 \Rightarrow a = -0.1$

Equation is $y = -0.1(x - 10)(x + 10)$ OR $y = -0.1x^2 + 10$ **[1]**

Domain: $-10 \leq x \leq 10$ **[1]**

Alternatively, students could substitute the three points $(-10, 0)$, $(0, 10)$ and $(10, 0)$ into $y = ax^2 + bx + c$. This would give three simultaneous equations to solve, giving $a = -0.1$, $b = 0$ and $c = 10$ and hence $y = -0.1x^2 + 10$.

d As the y-axis is an axis of symmetry of the seat, a diameter of the circle will lie on the y-axis and therefore the centre of the circle will lie on the y-axis **[1]**

Let the centre of the circle be $C\,(0, c)$, which will be above the seat **[1]**

The radius (along the y-axis) will be of length $-9 - c$. **[1]**

The radius also joins $C\,(0, c)$ and $(3, -8)$

Length of this radius $= \sqrt{(0-3)^2 + (c-(-8))^2} = \sqrt{9 + (c+8)^2}$ **[1]**

These are equal.

$$\sqrt{9 + (c+8)^2} = -9 - c$$ **[1]**

$$9 + c^2 + 16c + 64 = 81 + 18c + c^2$$ **[1]**

$$-2c = 8$$

$$c = -4$$ **[1]**

Looking at the y-axis, radius = 5 units (from -4 to -9) **[1]**

Students may also deduce this by trial and error, but must include hypotenuse calculation. They may observe 3-4-5 triangles, which is acceptable.

Equation: $x^2 + (y+4)^2 = 25$ **[1]**

Domain: $-3 \leq x \leq 3$ Range: $-9 \leq y \leq -8$ **[1]**

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e Start with $y = -kx^2(x - a)(x - b) - c$.

When $x = 0$, $y = -10 \Rightarrow -10 = -k \times 0^2(0 - a)(0 - b) - c \Rightarrow c = 10$. **[1]**

Thinking: This is the graph $y = -kx^2(x - a)(x - b)$ translated 10 units down, so the end-points are like zeros.

When $x = 10$, $y = -10 \Rightarrow -10 = -k \times (-10)^2(10 - a)(10 - b) - 10$ **[1]**

$\Rightarrow -k \times (-10)^2(10 - a)(10 - b) = 0 \Rightarrow a = 10$ or $b = 10$ (since k $\neq$ 0) **[1]**

It does not matter which one we choose, so let $a = 10$ **[1]**

(It is of course acceptable if students make $b = 10$ instead.)

When $x = -10$, $y = -10 \Rightarrow -10 = -k \times 10^2(-10 - 10)(-10 - b) - 10$

$\Rightarrow -k \times 10^2(-20)(-10 - b) = 0 \Rightarrow b = -10$ (since $k \neq 0$) **[1]**

We now have $y = -kx^2(x - 10)(x + 10) - 10$

The most obvious point to use is $(7, -7\frac{1}{2})$.

$-k \times 7^2 (7 - 10)(7 + 10) - 10 = -7\frac{1}{2}$ **[2]**

$$-49k(-3)(17) = 2\frac{1}{2}$$

$$k = \frac{2.5}{2499} = \frac{25}{24990} = \frac{5}{4998}$$ **[1]**

Exact equation: $y = \dfrac{-5x^2(x - 10)(x + 10)}{4998} - 10$ **[1]**

Students will observe that 4998 is very close to 5000 so for the garden the equation can be approximated to

$y = \dfrac{-x^2(x - 10)(x + 10)}{1000} - 10$ or $y = -0.001x^2(x - 10)(x + 10) - 10$ **[1]**

Substituting $x = 7$ to this equation gives $y = -7.501$.

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